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Java SE 27 & JDK 27

Class AbstractSelector

java.lang.Object
    java.nio.channels.Selector
        java.nio.channels.spi.AbstractSelector

All Implemented Interfaces:

Closeable, AutoCloseable

```
public abstract class AbstractSelector
extends Selector
```

Base implementation class for selectors.

This class encapsulates the low-level machinery required to implement the interruption of selection operations. A concrete selector class must invoke the `begin` and `end` methods before and after, respectively, invoking an I/O operation that might block indefinitely. In order to ensure that the `end` method is always invoked, these methods should be used within a `try` ... `finally` block:

```
    try {
        begin();
        // Perform blocking I/O operation here
        ...
    } finally {
        end();
    }
```

This class also defines methods for maintaining a selector's cancelled-key set and for removing a key from its channel's key set, and declares the abstract `register` method that is invoked by a selectable channel's `register` method in order to perform the actual work of registering a channel.

Since:

1.4

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	**AbstractSelector** (SelectorProvider provid	Initializes a new instance of this class.

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
protected final void	**begin**()	Marks the beginning of an I/O operation that might block indefinitely.
protected final Set<SelectionKey>	**cancelledKeys**()	Retrieves this selector's cancelled-key set.
final void	**close**()	Closes this selector.
protected final void	**deregister**(AbstractSelectionKey ke	Removes the given key from its channel's key set.
protected final void	**end**()	Marks the end of an I/O operation that might block indefinitely.
protected abstract void	**implCloseSelector**()	Closes this selector.
final boolean	**isOpen**()	Tells whether or not this selector is open.
final SelectorProvider	**provider**()	Returns the provider that created this channel.
protected abstract SelectionKey	**register**(AbstractSelectableChann int ops, Object att)	Registers the given channel with this selector.

Methods declared in class Selector

keys, open, select, select, select, select, selectedKeys, selectNow, selectNow, wakeup

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

AbstractSelector

```
protected AbstractSelector(SelectorProvider provider)
```

Initializes a new instance of this class.

Parameters:

`provider` - The provider that created this selector

Method Details

close

```
public final void close()
                throws IOException
```

Closes this selector.

If the selector has already been closed then this method returns immediately. Otherwise it marks the selector as closed and then invokes the `implCloseSelector` method in order to complete the close operation.

Specified by:

`close` in interface `AutoCloseable`

Specified by:

`close` in interface `Closeable`

Specified by:

`close` in class `Selector`

Throws:

`IOException` - If an I/O error occurs

implCloseSelector

```
protected abstract void implCloseSelector()
                                    throws IOException
```

Closes this selector.

This method is invoked by the `close` method in order to perform the actual work of closing the selector. This method is only invoked if the selector has not yet been closed, and it is never invoked more than once.

An implementation of this method must arrange for any other thread that is blocked in a selection operation upon this selector to return immediately as if by invoking the `wakeup` method.

Throws:

`IOException` - If an I/O error occurs while closing the selector

isOpen

```
public final boolean isOpen()
```

Description copied from class: `Selector`

Tells whether or not this selector is open.

Specified by:

`isOpen` in class `Selector`

Returns:

`true` if, and only if, this selector is open

provider

```
public final SelectorProvider provider()
```

Returns the provider that created this channel.

Specified by:

`provider` in class `Selector`

Returns:

The provider that created this channel

cancelledKeys

```
protected final Set<SelectionKey> cancelledKeys()
```

Retrieves this selector's cancelled-key set.

This set should only be used while synchronized upon it.

Returns:

The cancelled-key set

register

```
protected abstract SelectionKey register
(AbstractSelectableChannel ch,
 int ops,
 Object att)
```

Registers the given channel with this selector.

This method is invoked by a channel's `register` method in order to perform the actual work of registering the channel with this selector.

Parameters:

`ch` - The channel to be registered

`ops` - The initial interest set, which must be valid

`att` - The initial attachment for the resulting key

Returns:

A new key representing the registration of the given channel with this selector

deregister

```
protected final void deregister(AbstractSelectionKey key)
```

Removes the given key from its channel's key set.

This method must be invoked by the selector for each channel that it deregisters.

Parameters:

`key` - The selection key to be removed

begin

```
protected final void begin()
```

Marks the beginning of an I/O operation that might block indefinitely.

This method should be invoked in tandem with the `end` method, using a `try` ... `finally` block as shown above, in order to implement interruption for this selector.

Invoking this method arranges for the selector's `wakeup` method to be invoked if a thread's `interrupt` method is invoked while the thread is blocked in an I/O operation upon the selector.

end

```
protected final void end()
```

Marks the end of an I/O operation that might block indefinitely.

This method should be invoked in tandem with the `begin` method, using a `try` ... `finally` block as shown above, in order to implement interruption for this selector.

Report a bug or suggest an enhancement
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